Chapter 9 Earth in space

EXTRA WORKSHEET 1: Missions to the Moon

VERBAL/LINGUISTIC LOGICAL/MATHEMATICAL SCIENCE AS A HUMAN ENDEAVOUR

Name:_____Class:_____Date: _____

The Moon is the closest celestial body to Earth and so it's not surprising that it was the first to be explored. Apart from Earth, the Moon remains the only body in space that humans have walked upon.

Early exploration

USA and the USSR (Russia) sent roughly 40 missions to the Moon between 1959 and 1969.

Timeline of some important early lunar missions			
Year	**Mission**	**Country**	**What happened**
1959	*Luna 1*	USSR (Russia)	First unmanned craft to reach the Moon. Deliberately crashed into surface.
	Pioneer 4	USA	First US mission to the Moon. Flew by and on into space.
	Luna 3	USSR (Russia)	Passed by and beamed back to Earth first-ever images of the far side.
1966	*Lunar 9*	USSR (Russia)	First soft landing on the surface. Proved that the lunar surface could support a spacecraft.
1967	*Surveyor 3*	USA	Landed and took the first samples of lunar soil.

The Apollo missions

Between 1969 and 1972, six Apollo missions successfully landed two men each on the Moon's surface and brought them safely back to Earth. The Apollo program, however, started with disaster: a fire raced through the command module of *Apollo 1* during a test of launch procedures, killing all three astronauts on board. *Apollo 13* also suffered disaster: an explosion and multiple system failures caused its Moon landing to be abandoned and it was under continued threat of further disaster on its way back to Earth.

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The first human to walk on the Moon was Neil Armstrong, followed shortly by his crewmate Edwin (Buzz) Aldrin. They spent roughly a day on the lunar surface, venturing outside of their lunar module for a total of 2½ hours. In 1972, Eugene Cernan was the last person to walk on the Moon. Since then, the furthest a human has gone into space is into orbit around Earth, a few hundred kilometres above its surface.

Timeline of selected Apollo missions			
Date	**Mission**	**Country**	**What happened**
January 1967	*Apollo 1*	USA	Fire on launch pad kills three astronauts on board.
December 1968	*Apollo 8*	USA	First Apollo mission to travel to the Moon, go into orbit and return.
July 1969	*Apollo 11*	USA	First landing of humans on the Moon, on Sea of Tranquillity.
April 1970	*Apollo 13*	USA	Explosions onboard. Mission aborted. Orbits Moon and returns to Earth.
December 1972	*Apollo 17*	USA	Last manned Moon landing. Landed on Taurus-Littrow. Third Apollo mission to use LRV (Moon buggy) on the lunar surface.

The future

Only a few missions (all unmanned) have been launched towards the Moon since the final Apollo mission in 1972. The last few years, however, have seen Japan, China, India and the European Space Agency join in the exploration of the Moon.

Timeline of recent lunar missions			
Year	**Mission**	**Country**	**What happened**
2004	*Smart-1* (European Space Agency)	EEC (European Economic Community)	First European lunar mission
2007	*Chang-e 1*	China	First Chinese lunar mission
2008	*Chandrayaan-1*	India	First Indian lunar mission
2009	*Kaguya*	Japan	First Japanese lunar mission

The USA plans to establish a base on the Moon by 2020 as a stepping stone to a manned mission to Mars. China plans a manned lunar landing sometime between 2020 and 2025.

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1 **State** whether the following statements are true or false.

a Apart from Earth, the Moon is the only body in space to be visited personally by humans.

_____

b The space shuttle and current rockets have taken humans as far as the Moon in recent years.

_____

2 **State** the year of:

a the first unmanned mission to the Moon.

_____

b the first manned landing on the Moon.

_____

c the last time humans visited the Moon in person.

_____

3 **Name** the Apollo mission that:

a resulted in the deaths of three astronauts.

_____

b was the first to land humans on the Moon.

_____

c was the last to land on the Moon.

_____

d was aborted because of an explosion and system failures.

_____

4 **Name** the two astronauts who first walked on the lunar surface.

_____

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5 *Apollo 15*, *16* and *17* missions used a car-like Lunar Rover Vehicle (LRV or Moon buggy) to explore the surface of the Moon. All three vehicles were left behind. **Propose** reasons why.

_____

_____

_____

6 **Name** the countries that:

a were most active in early Moon exploration.

_____

b have only recently begun to explore the Moon.

_____

7 **Explain** why the USA is planning to construct a Moon base by 2020.

_____

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